

Project Initialization and Planning Phase

Date	28 June 2026
Team ID	xxxxxx
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform learner support using deep learning and generative AI, boosting empathy and responsiveness in academic help. It tackles the lack of emotion-aware feedback in self-study, promising better engagement, reduced frustration, and more confident learners. Key features include dual deep-learning emotion classifiers and Gemini-generated personalized guidance.

Project Overview	
Objective	The primary objective is to help learners receive emotion-aware, personalized study support by analyzing free-text descriptions of their challenges using deep learning emotion classifiers and generative AI guidance.
Scope	The project ingests a student's free-text problem description, predicts the underlying emotion using BiLSTM and BERT models enhanced with rule-based keywords, generates tailored tips via Gemini, logs interactions to CSV, and visualizes emotion trends through an analytics dashboard.
Problem Statement	
Description	Many students studying independently, and the instructors supporting them, struggle to recognize and respond to emotional states like confusion or frustration in real time, leading to slower resolution of academic challenges and reduced support quality.
Impact	Solving this problem improves learner support quality, reduces manual triage effort for educators, and increases student confidence and engagement in self-paced learning.
Proposed Solution	
Approach	Employ deep learning (BiLSTM, BERT), rule-based keyword enhancement, Gemini generative AI, and Streamlit to classify a learner's emotion from text and generate personalized, empathetic guidance.
Key Features	- Dual deep learning emotion classifiers (BiLSTM & BERT). - Rule-based keyword enhancement for improved accuracy. - Gemini-generated personalized tips & encouragement. - Mixed-emotion detection. - Interaction logging to CSV. - Analytics dashboard for emotion trend visualization. - Interactive Streamlit interface.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5 Processor (or above)
Memory	RAM specifications	8 GB (16 GB recommended)
Storage	Disk space for data, models, and logs	2 GB free disk space (minimum)
Software		
Frameworks	Python frameworks	Streamlit, PyTorch
Libraries	Additional libraries	Transformers, NumPy, Pandas, ReportLab, Gemini API SDK
Development Environment	IDE	Jupyter Notebook, VS Code
Data		
Data	Source, size, format	User-provided free-text problem descriptions; CSV files (interactions.csv) for interaction and feedback logs